



First Name: _____

Family Name: _____

Student Number: _____

Diploma of Engineering
University Entrance Program
Foundation Studies Program

DPEN021 – Enabling Mathematics A

Final Exam: Sample

This exam represents 48% of the total marks for the subject

Time allowed:	2 hours
Items permitted in exam:	UOW approved calculator
Aids supplied for exam:	Formula Sheet

INSTRUCTIONS TO STUDENTS

1. Write in blue or black pens only. Pencils are permitted for diagrams and graphs.
2. All questions are to be attempted.
3. All notation is as given in lectures.
4. Show all necessary WORKING and reasoning in the spaces provided.
5. Write the letter (A, B, C, D, or E) corresponding to the correct answer of each question in the box provided.
6. You must achieve 40% in the Final Exam. If this requirement is not met, your final mark for the subject will be a Technical Fail
7. Total number of questions: 28
 - Questions 1-12 are multiple choice
 - Each question is worth 1 mark
 - Marks are awarded for the answer only.
 - Questions 13-28 are long answer questions
 - The value of each question is indicated.
 - Part marks may be awarded.

No examination material may be removed from the examination venue.

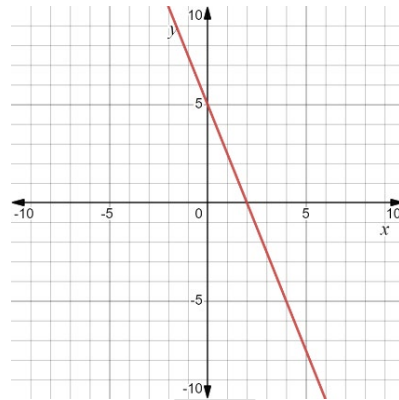
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Questions 1-14 are multiple choice. Show all necessary *WORKING* and reasoning in the spaces provided. Write the letter (A, B, C, D, or E) corresponding to the correct answer of each question in the box provided.

Question 1 [Straight Lines]

The equation of this straight line is:

- (A) $2y + 5x = 10$
- (B) $5x + 2y = 1$
- (C) $2x + 5y = 1$
- (D) $y = 2x + 5$
- (E) None of the above



Q1

Question 2 [Straight Lines]

The graph of $x = 2$ would be

- (A) A sloping line
- (B) A horizontal line
- (C) A vertical line
- (D) A curved line
- (E) None of the above

Q2

Question 3 [Straight Lines]

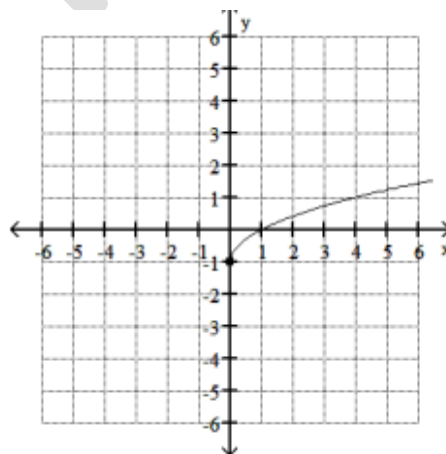
The gradient of the line passing through the points $(-2a, -a)$ and $(-10, -5)$ is:

- (A) 2
- (B) $-\frac{1}{2}$
- (C) -2
- (D) $\frac{1}{2}$
- (E) a

Q3**Question 4 [Domain and Range]**

Find the domain and range of the graph.

- (A) Domain: $[0, \infty)$; Range $(-\infty, \infty)$
- (B) Domain: $(-\infty, \infty)$; Range $[-1, \infty)$
- (C) Domain: $[0, \infty)$; Range $[-1, \infty)$
- (D) Domain: $[0, \infty)$; Range $[0, \infty)$
- (E) Domain: $[0, \infty]$; Range $(0, \infty)$

**Q4**

Question 5 [Piecemeal Functions/Composite Functions]

Given the piecemeal function $f(x) = \begin{cases} 3x+2 & x < -1 \\ x^2 & -1 \leq x < 2 \\ \sqrt{x} & x \geq 2 \end{cases}$.

Calculate $f(-3) + f(0) + 2f(4)$.

- (A) 0
- (B) -1
- (C) -5
- (D) 5
- (E) -3

Q5

Question 6 [Quadratic Functions]

The graph of the function $y = 2x^2 + 8x + 6$ has:

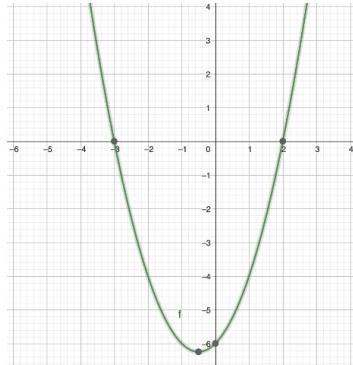
- (A) a minimum turning point at (2, -2)
- (B) a minimum turning point at (-2, 2)
- (C) a minimum turning point at (-2, -2)
- (D) a maximum turning point at (-2, -2)
- (E) a maximum turning point at (2, -2)

Q6

Question 7 [Quadratic Functions]

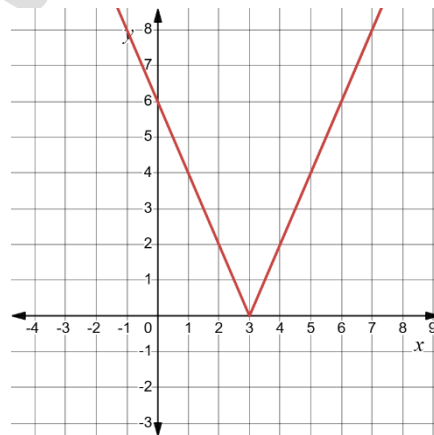
Which equation represents the following graph:

- (A) $y = (x + 3)(x + 2)$
- (B) $y = (x - 3)(x + 2)$
- (C) $y = (x + 3)(x - 2)$
- (D) $y = (x - 3)(x - 2)$
- (E) $y = (x + 3)^2$

**Q7****Question 8 [Absolute Value Functions]**

Which equation describes the following graph?

- (A) $y = |6 + 2x|$
- (B) $y = |6 - 2x|$
- (C) $y = |2x| + 6$
- (D) $y = 6 - |2x|$
- (E) None of the above

**Q8**

Question 9 [Circles]

What is the equation of the circle with centre $(-1, 2)$ and radius 3 units?

- (A) $(x-1)^2 + (y-2)^2 = 9$
- (B) $(x+1)^2 + (y+2)^2 = 9$
- (C) $(x+1)^2 + (y-2)^2 = 9$
- (D) $(x+1)^2 + (y+2)^2 = 3$
- (E) $(x-1)^2 + (y-2)^2 = 3$

Q9**Question 10 *Logarithm and Exponential Functions***

Expressed as a single logarithm $\log_a x - \log_a y + 3 \log_a z$ equals.

- (A) $\log_a \frac{xz^3}{y}$
- (B) $\log_a \frac{3xz}{y}$
- (C) $\log_a \frac{x}{z^3 y}$
- (D) $\log_a \frac{x}{3yz}$
- (E) None of these

Q10

Question 11 [Logarithm and Exponential Functions]

Solve the equation $\log_{10} x - \log_{10} (x-1) = 1$ and select the correct solution from the options below.

- (A) $x = \frac{9}{10}$
- (B) $x = \frac{e}{e-1}$
- (C) $x = \frac{10}{9}$
- (D) $x = 1$
- (E) None of these

Q11

Question 12 [Logarithms and Exponentials]

The graph of $g(x)$ is obtained by shifting the graph of $f(x) = e^x$ one unit to the left and two units down. Determine the equation of $g(x)$.

- (A) $g(x) = e^{x-1} - 2$
- (B) $g(x) = e^{x+1} + 2$
- (C) $g(x) = e^{x+2} - 1$
- (D) $g(x) = e^{x+1} - 2$
- (E) $g(x) = e^{x-2} + 1$

Q12

Questions 15-30 are long answer questions. Show all necessary WORKING and reasoning in the spaces provided. The value of each question is indicated. Part marks may be awarded.

Question 15 [Straight Lines] (2 marks)

Find the equation of the line passing through the points $(-2,1)$ and $(1,6)$. Leave your answer in gradient-intercept form.

Question 16 [Straight Lines] (2 marks)

The line with equation $y + 3x = 10$ passes through the point $(2, a)$. Find the value of a .

Question 17 [Straight Lines] (3 marks)

Find the equation of the straight line that passes through the point (2, 2) and is perpendicular to the line $y - 2x = 1$. Leave your answer in general form.

Question 18 [Simultaneous Equations] (2 marks)

Solve the simultaneous linear equations, showing all working.

$$2x + 3y = 19$$

$$4x - y = 3$$

Question 19 [Domain and Range] (2 marks)

Determine the domain and range of the function $y = \frac{1}{2x+1}$, leave your answer in interval notation.

Question 20 [Odd and Even Functions] (2 marks)

Is the function $f(x) = 2x^4 - x^2$ odd, even or neither? Show all working to justify your answer.

Question 21 [Composite Functions/Piecemeal Functions] (2 marks)

Given $g(x) = \frac{1}{x^2 + 2x}$ and $h(x) = 3x$. Evaluate $g(h(-2))$.

Question 22 [Quadratic Functions] (2 marks)

Find all values of a for which the equation $x^2 + ax + 36 = 0$ has one real root.

Question 23 [Quadratic Functions] (3 marks)

Graph the function $y = -x^2 + 16$. Calculate and show all important features including the intercepts and vertex.

Question 24 [Absolute Value Functions] (2 marks)

Sketch the graph of $y = |x + 1| + 2$. Show all important features.

Question 25 [Circles] (3 marks)

Find the centre and the radius of the circle $x^2 + y^2 - 2y = 3$.

Question 26 [Circles] (2 marks)

Sketch the above circle $x^2 + y^2 - 2y = 3$ and find the intercepts.

Question 27 [Logarithm and Exponential Functions] (2 marks)

Evaluate $\log_{\sqrt{2}} 32$.

Question 28 [Logarithm and Exponential Functions] (3 marks)

Solve the equation $3e^{x-5} + 1 = 10$ for x , giving the answer to 3 decimal places.

Question 29 [Logarithm and Exponential Functions] (3 marks)

Sketch the graph of $y = \log_e (x - 2)$. Find the x -intercept and write the equation of the vertical asymptote.

Question 30 [Logarithm and Exponential Functions] (2 marks)

Sketch the graph of $y = e^x + 4$. Find the y -intercept and write the equation of the horizontal asymptote.

END OF EXAM

Formula Sheet

Straight Line

Gradient: $m = \frac{y_2 - y_1}{x_2 - x_1}$

Gradient-Intercept form: $y = mx + b$

General form: $Ax + By + C = 0$

Point-Gradient form: $y - y_1 = m(x - x_1)$

Parallel lines: $m_1 = m_2$

Perpendicular lines: $m_1 \times m_2 = -1$

Functions

Odd function: $f(-x) = -f(x)$

Even function: $f(-x) = f(x)$

Composite functions: $(f \circ g)(x) = f(g(x))$

Quadratic Functions

Equation of quadratic: $ax^2 + bx + c = 0$

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Discriminate: $\Delta = b^2 - 4ac$

Axis of symmetry: $x = -\frac{b}{2a}$

Vertex: $\left(-\frac{b}{2a}, f\left(-\frac{b}{2a}\right)\right)$

Equation of a circle

$$(x - a)^2 + (y - b)^2 = r^2$$

Rules for Logarithms

$x = \log_b N$ if and only if $b^x = N$

$$\log_b xy = \log_b x + \log_b y$$

$$\log_b \left(\frac{x}{y}\right) = \log_b x - \log_b y$$

$$\log_b x^p = p \log_b x$$

$$\log_b x = \frac{\log_a x}{\log_a b}$$

$$\log_b b = 1$$

$$\log_b b^x = x$$

$$\log_b 1 = 0$$

$$b^{\log_b x} = x$$